Department of Applied Computing and Artificial Intelligence

Faculty of Computing

SECR1213 NETWORK COMMUNICATIONS

Project Task 4

Making the Connections – LAN and WAN (10%)

Group of *Power Rangers*

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Submission Date: 28th January 2025

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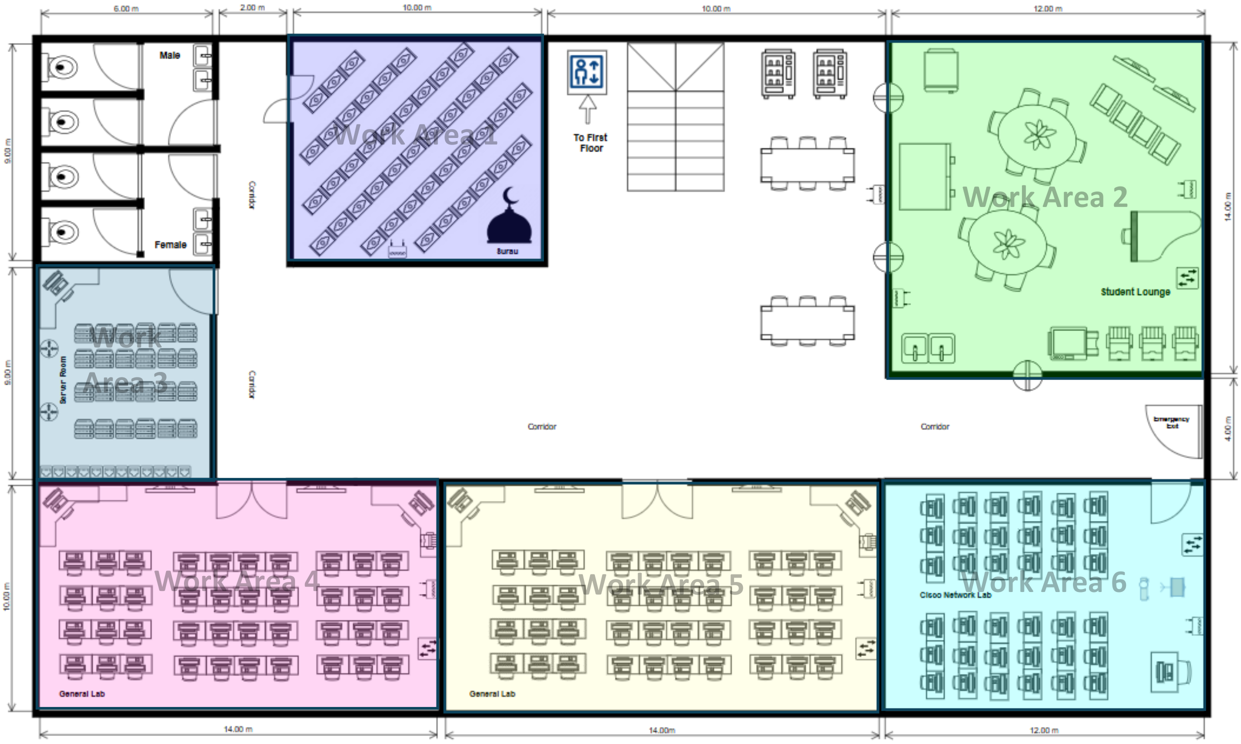
1.0 Identify the work areas

1.1 Ground Floor



Work Area 1 (Surau) located in the middle top section, the surau is a peaceful space for prayers and spiritual activities. It provides a quiet and respectful setting for personal religious practices.

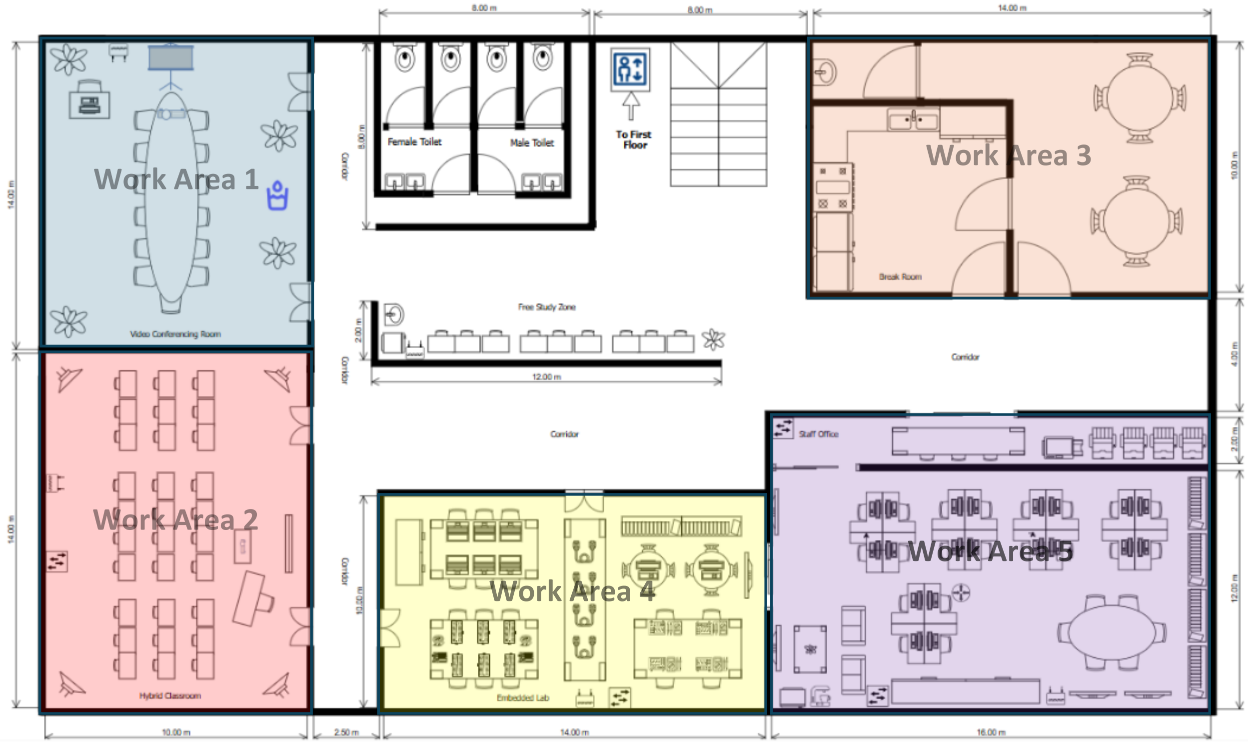
Work Area 2 (Student Lounge) situated in the top right corner, the student lounge is a cosy area for students to relax, socialize, and work together. It is designed for informal gatherings and leisure activities.

Work Area 3 (Server Room) situated in the middle left, the server room securely stores critical data. It has two PC workstations and three servers, all connected for smooth data transfer. The room is equipped with strong monitoring and security features.

Work Areas 4 & 5 (General Labs) found in the bottom left and middle sections, these general labs are multipurpose spaces for academic and technical activities. They support a wide range of learning and practical needs.

Work Area 6 (Cisco Network Lab) located in the bottom right corner; the Cisco network lab is designed for networking practice. It includes a workstation, wireless access point, and two printers to help with hands-on learning and projects.

1.2 First Floor



Work Area 1 (Video Conference Room) located in the top left, this room is for faculty to conduct virtual meetings, lectures, and collaborative work. It has high-quality audio-visual tools for screen sharing and remote interaction, making it ideal for blended learning, administrative tasks, and hosting guest speakers without travel.

Work Area 2 (Hybrid Classroom) positioned in the bottom left, this classroom supports both in-person and remote learning. It includes advanced technology like cameras, microphones, and screen-sharing tools, ensuring a smooth learning experience for all students and promoting inclusivity.

Work Area 3 (Break Room) situated in the top right, this room is a relaxing space for faculty and staff. It offers comfortable seating and basic amenities, encouraging informal interactions and providing a break from work, supporting a balanced work environment.

Work Area 4 (Embedded Lab) situated in the middle bottom section, this lab focuses on developing and testing embedded systems. With 2 PC workstations, an access point, and a large TV screen, it offers hands-on experience in electronics, robotics, and microcontroller projects, preparing students for practical, innovative work.

Work Area 5 (Staff Office) located in the bottom right, this office provides workstations for faculty and staff to manage daily academic and administrative tasks. This makes it more productive in a functional and organised environment.

2.0 Overall Network Connection Diagram

2.1 Network Distribution Diagram

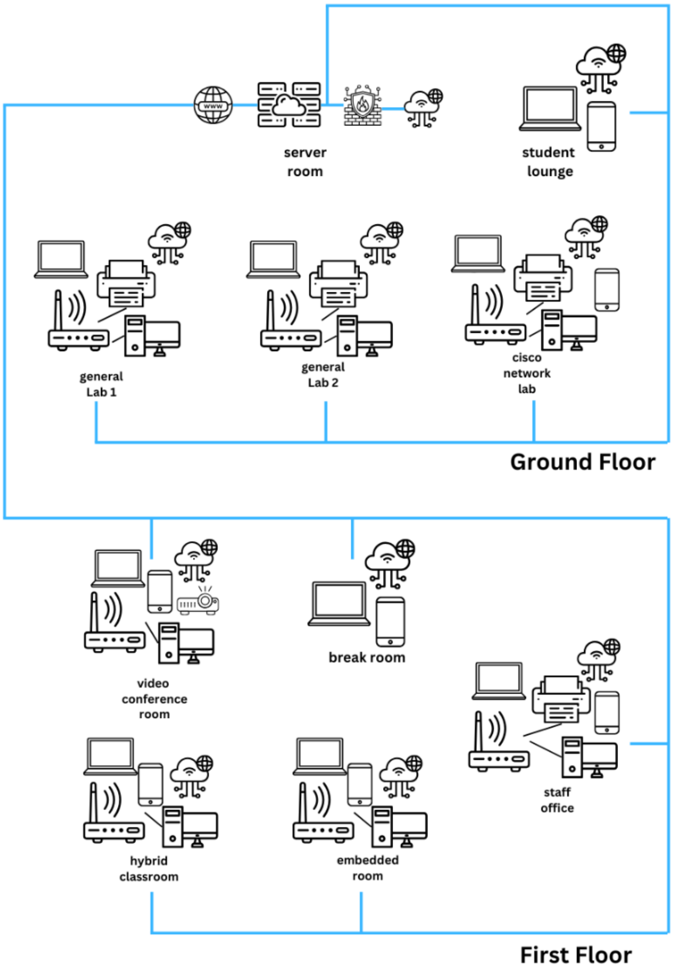


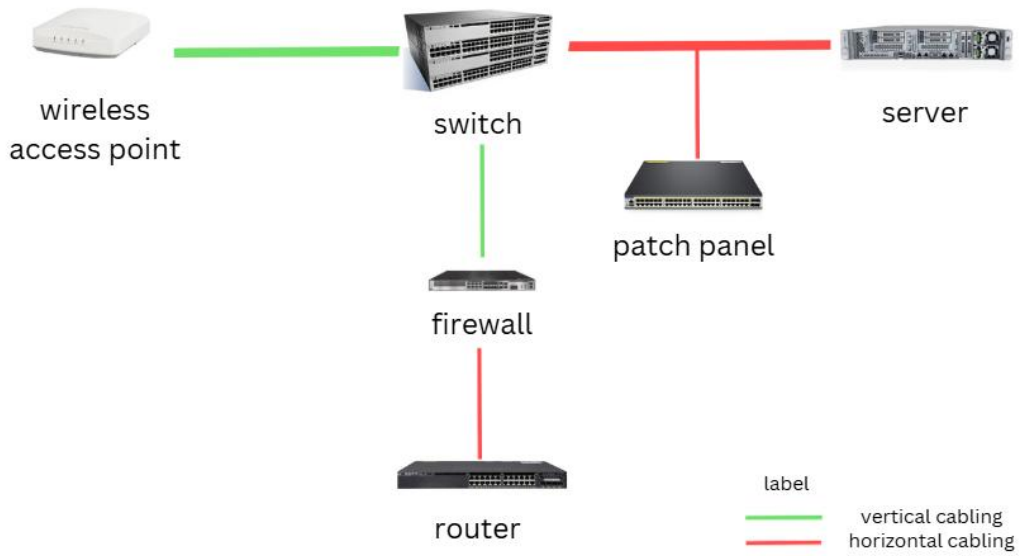
Figure 1 Network Distribution Diagram

Figure 1 shows the network layout for the building's ground and first floors. Each key area, such as the hybrid classroom, video conference room, embedded lab, staff office, and break room, will have a wireless access point (WAP) and a network switch for reliable internet and smooth local communication. High-traffic areas on the ground floor, including the two general labs, Cisco network lab, and student lounge, will also use switches to handle heavy data flow efficiently.

The switches will be interconnected to form subnets, which improve performance and reduce pressure on the central router. Extra ports on the switches are available for connecting more devices, allowing for easy future upgrades. This design ensures fast data transmission, low network delays, and dependable connectivity for all users and devices in the building.

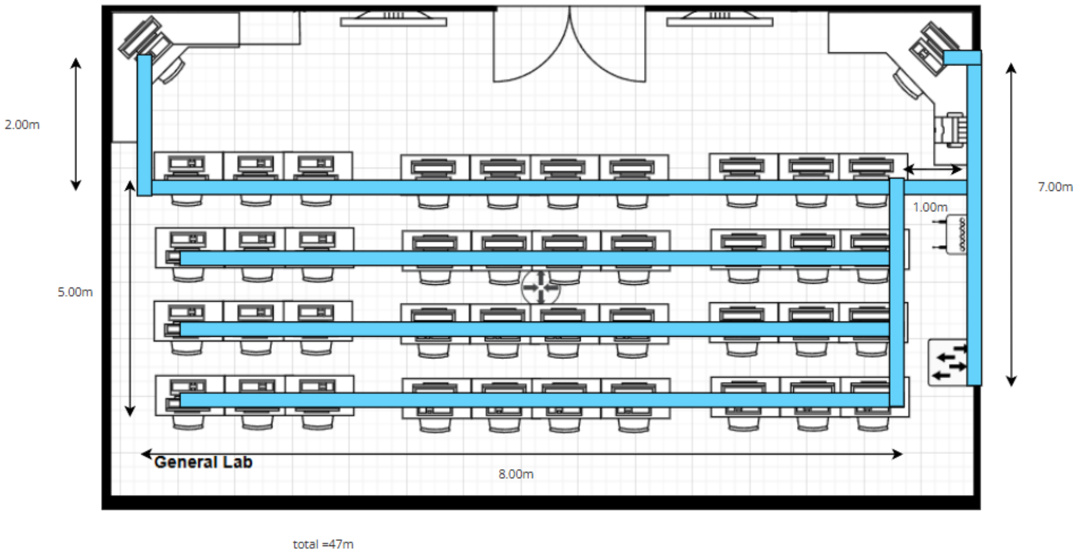
2.1.1[Ground Floor] Server Room

As all network devices are placed in the server room, this is the closer look in the server room:

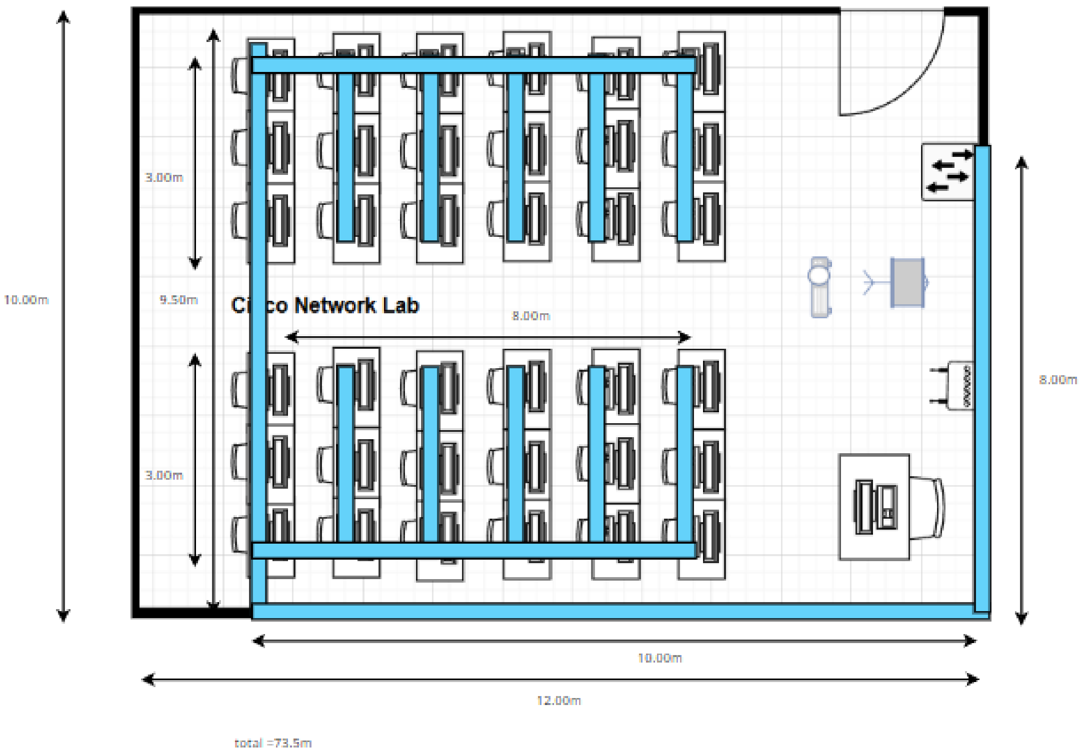


The diagram illustrates a network architecture with key components interconnected using both vertical and horizontal cabling. A wireless access point (WAP) connects to the switch via vertical cabling to provide wireless connectivity for devices. The switch serves as the central hub, connecting multiple components, including the patch panel, server, and firewall. The patch panel organizes and manages cable connections, while the server provides resources such as file storage and application hosting. The switch is connected to the server and patch panel via horizontal cabling for data distribution. A firewall sits between the switch and the router, monitoring and securing traffic to protect the network from external threats. The router connects the local network to the internet, enabling external communication. Vertical cabling connects the router, firewall, and WAP, while horizontal cabling links the switch, server, and patch panel, ensuring efficient and secure communication throughout the network.

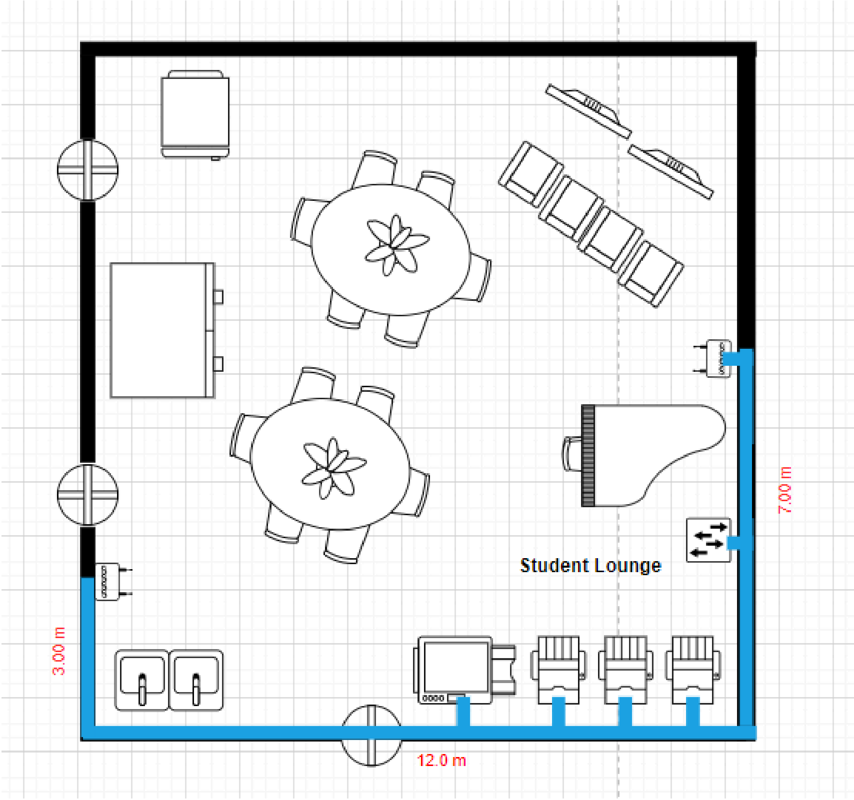
2.1.2 [Ground Floor] General Purpose Lab



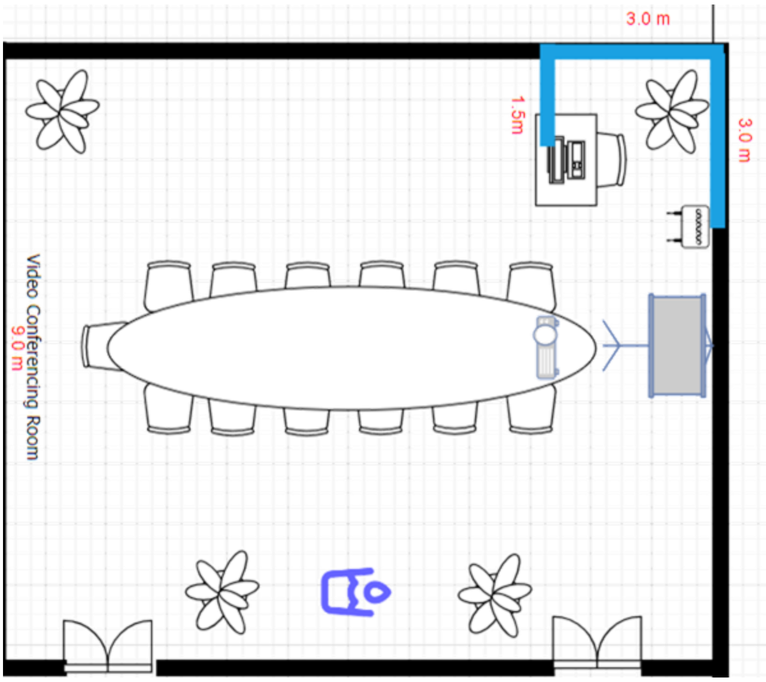
2.1.3 [Ground Floor] Cisco Network Lab



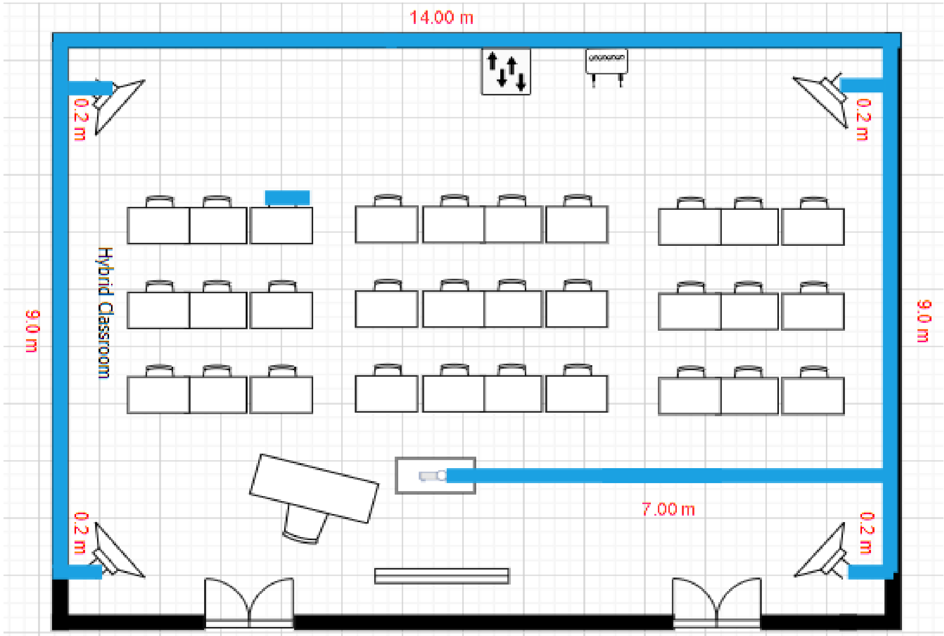
2.1.4 [Ground Floor] Student Lounge



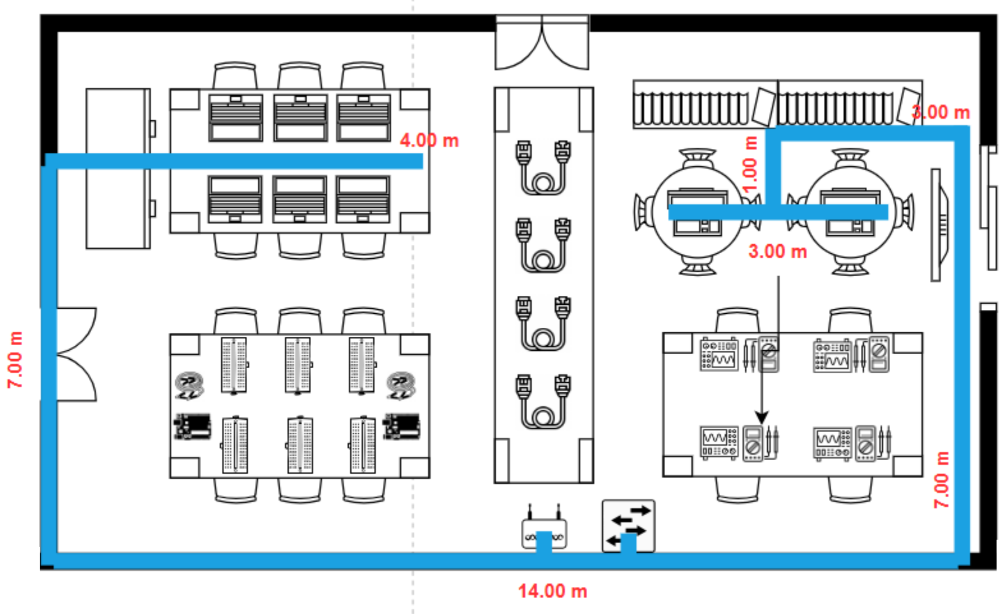
2.1.5 [First Floor] Video Conferencing Room



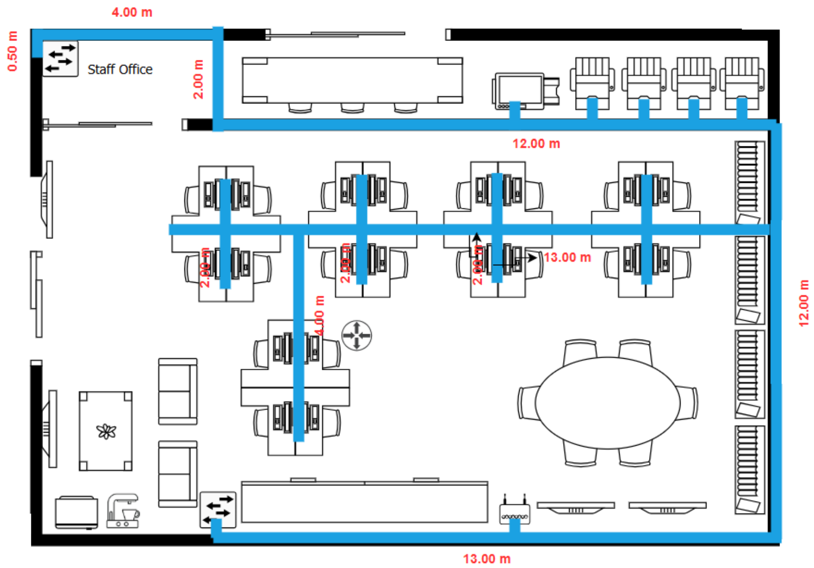
2.1.6 [First Floor] Hybrid Classroom



2.1.7 [First Floor] Embedded Lab



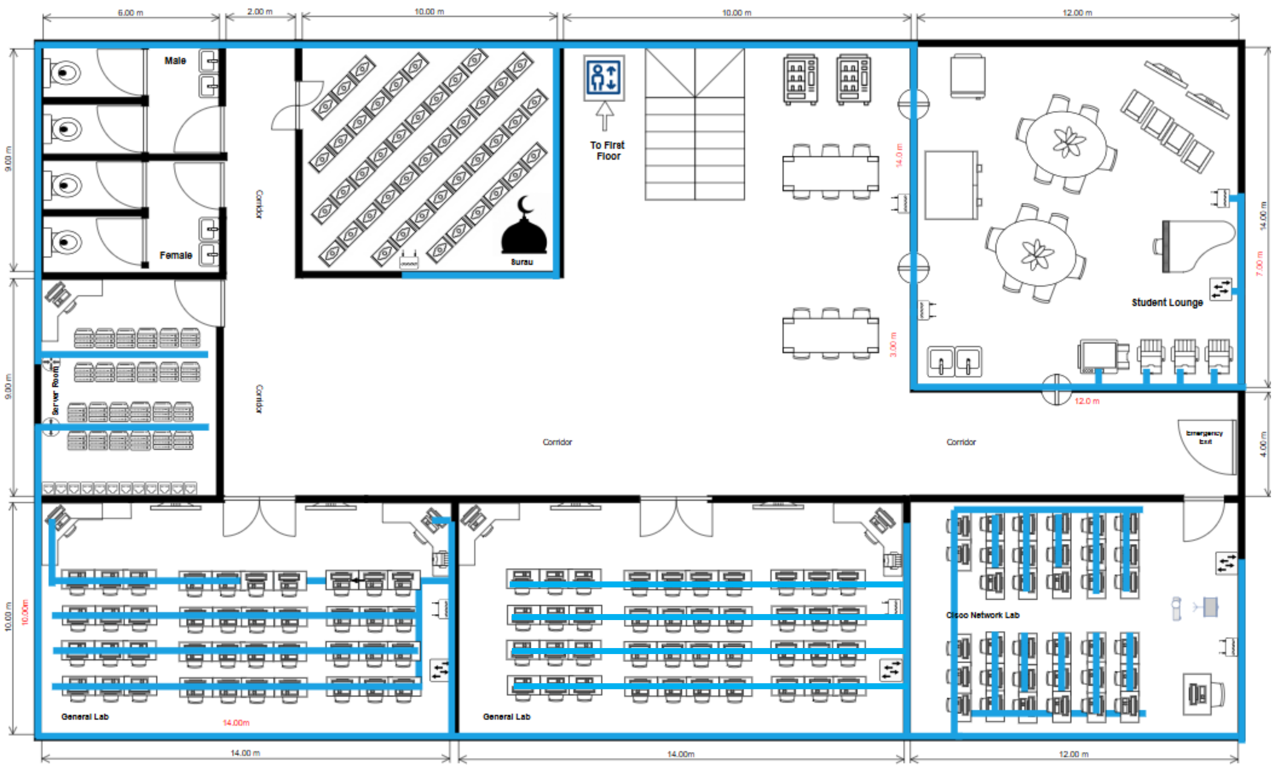
2.1.8 [First Floor] Staff Office



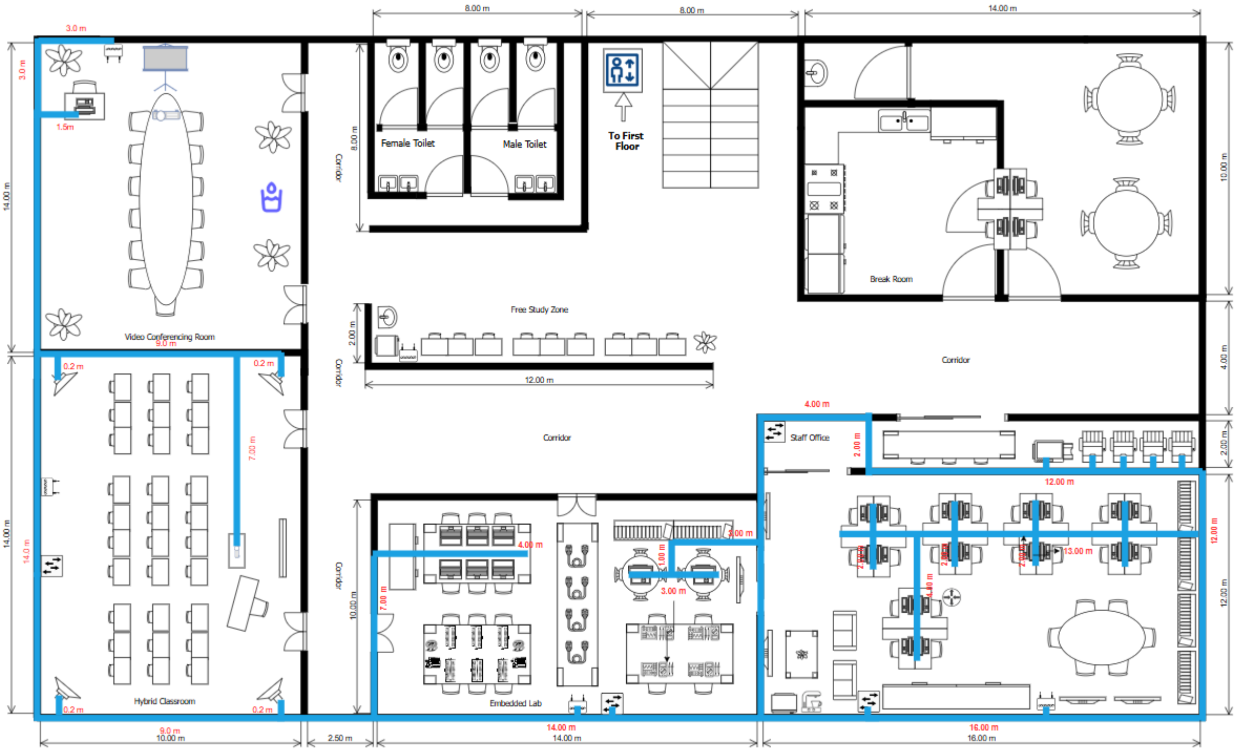
3.0 Cables & Connections

3.1 Floor Plan

3.1.1 Ground Floor Plan



3.1.2 First Floor Plan



Figures 3.1.1 and 3.1.2 present detailed floor plans shows the proposed network layout for the faculty building, illustrating the connections between individual rooms. The network is centralised in the server room, serving as the core for managing internet and intranet traffic. Each room is connected to this central hub via a backbone connection, ensuring reliable and efficient data transmission throughout the building. Additionally, the diagrams demonstrate the strategic placement of wireless access points across both the ground and first floors, providing comprehensive wireless network coverage to support seamless connectivity for all users within the facility.

3.1.3 Number of Connections, Path Cords, and Switch Ports Needed

Area	Number of Switch Ports Needed
General Lab 1	30
General Lab 2	30
Hybrid classroom	5
Video Conferencing Room	1
Server Room	8
Embedded Lab	8
Cisco Networking Lab	30
Staff Office	15
Surau	1
Break Room	1
Student Lounge	4
Total	**133**

Table 1: Number of Switch Port needed

Area	Number of Patch Cord Needed
General Lab 1	30
General Lab 2	30
Hybrid classroom	5
Video Conferencing Room	1
Server Room	8
Embedded Lab	8
Cisco Networking Lab	30
Staff Office	15
Surau	1
Break Room	1
Student Lounge	4
Total	**133**

Table 2: Number of Patch Cords needed

3.1.4 Identify Cable Types and Length

Area	Length(m)		Total Length (m)	Total Price of Cable Used (RM)
	Horizontal	Vertical		
Ground Floor				
General Purpose Lab	33.50	14.00	47.50	712.03
Cisco Network Lab	26.00	47.50	73.50	1,101.77
Student Lounge	12.50	12.00	24.50	367.26
Surau	16.00	9.50	25.50	382.25
		Total	**145.50**	**2,563.31**
First Floor				
Video Conferencing Room	3.00	4.50	7.50	112.43
Hybrid Classroom	21.80	18.00	39.80	596.60
Embedded Lab	24.00	16.00	41.00	614.59
Staff Office	29.00	30.00	59.00	943.41
		Total	**147.30**	**2,267.03**

Path	Total Length (m)	Total Price of Cable Used (RM)
Ground Floor		
Server Room to Surau	20.00	299.80
Surau to Student Lounge	10.00	149.90
Server Room to General Purpose Lab 1	3.00	44.97
Total	**33.00**	**494.67**
First Floor		
Hybrid Class to Embedded Lab	2.50	37.48
Total	**2.50**	**37.48**

	Total Length of Cable Used (m)	Total Price of Cable Used (RM)
Ground Floor	178.50	3057.98
First Floor	61.50	2304.51
Total	**240**	**5362.49**

The building's network design requires a total of **133 switch ports** and an equal number of **patch cords** to connect devices effectively. These ports are distributed across various areas based on their specific needs. For example, **General Labs 1 and 2** and the **Cisco Networking Lab** each require 30 ports due to their high device usage, while smaller areas like the **Surau** and **Break Room** need only 1 port each. Other spaces, such as the **Staff Office** and **Student Lounge**, require 15 and 4 ports, respectively, based on the number of devices expected. This distribution ensures adequate connectivity for all work and study environments.

The estimated total cable length required to connect all devices is **240 meters**, with a total cost of **RM 5,362.49**. On the **Ground Floor**, 145.5 meters of cable is required for areas like the **General Labs**, **Cisco Networking Lab**, and **Student Lounge**, costing **RM 3,057.98**. The **First Floor** requires 61.5 meters for spaces such as the **Video Conferencing Room**, **Hybrid Classroom**, and **Staff Office**, costing **RM 2,304.51**. Additional path cables of 33 meters are needed to link specific points on the Ground Floor, and 2.5 meters are required on the First Floor for direct connections, ensuring seamless connectivity across the building.

CAT6 Cables are used extensively for both horizontal and vertical cabling within the building. These cables support high-speed data transmission at speeds of up to **10 Gbps** and a bandwidth capacity of **550 MHz**, making them ideal for data-intensive activities such as cloud computing, coding, and large file transfers. They also support **Power over Ethernet (PoE)**, simplifying installation by combining power and data delivery in a single cable. Additionally, the design enhances durability and reduces electromagnetic interference, ensuring a stable and secure connection throughout the building.

Fibre Optic Cables are utilized for connecting routers in the **server room** and linking the network to the **Internet Service Provider (ISP)**. These cables transmit data using light, enabling extremely fast data speeds and minimal signal loss. They are particularly suited for long-distance connections and high-bandwidth applications, supporting speeds of up to **10 Gbps**. Fibre optics ensure efficient and reliable communication for core network connections, essential for the building's overall performance.

The combination of **CAT6** and **fibre optic cables** provides an optimal solution for the building's networking needs. CAT6 cables deliver high-speed, reliable internal connections, while fibre optic cables establish a strong external link to the ISP. This setup ensures smooth, efficient communication across all devices and areas, meeting the demands of both high-performance systems and everyday network usage.

To sum up, the total cost of designing this building amounts to RM1,187,402.27, in which RM1,182,039.78 is allocated for the required devices and RM5,362.49 for device cabling. With a total budget of RM5 million allocated for the building, the remaining RM3,812,597.73 can be distributed to cover key aspects such as construction, architectural design, materials, labour, utilities, permits, and other operational expenses.

Meeting Minutes #1

Date/Time	28th December 2024 10:00am	
Location	Online (Google Meet)	
Agenda	1. Review of work area details on both ground and first floors	
	2. Discussion of network distribution layout	
	3. Evaluation of required cables, connections, and cost estimation	
	4. Confirmation of server room setup, and labs' configurations	
Meeting Host	Elijah She Yu Sheng	
Attendance		
Name	**Time**	**Reason For Absence**
Elijah She Yu Sheng	10:00am	-
Lau Yan Kai	10:00am	-
Brendan Chia Yan Fei	10:00am	-
Chew Chiu Xian	10:00am	-

Minutes

No.	Item Discussed	Details	Person-In-Charge
1	Work Areas Identification	- Reviewed the details of work areas on both ground and first floors, including their purposes	Elijah
2	Network Distribution Diagram	- Discussed the network layout for the building, covering WAPs, switches, and subnet formation	Brendan
3	Server Room Design	- Reviewed the architecture, including placement of switches, patch panel, firewall, and router	Chiu Xian
4	Required Cables and Connections	- Identified the total number of switch ports, patch cords, and cable lengths for each area	Yan Kai

5	Cable Type and Cost Analysis	- Confirmed the use of CAT6 and fibre optic cables, along with cost estimation of RM 5,362.49	Yan Kai
6	General Labs and Cisco Lab Setup	- Discussed configuration of labs with necessary devices and connectivity for optimal performance	Chiu Xian
7	Hybrid Classroom and Embedded Lab Features	- Finalised tech requirements to support in-person and remote learning	Brendan
8	Staff Office and Break Room Connectivity	- Discussed ensuring reliable network access in these areas	Elijah
9	Wireless Access Point Placement	- Reviewed the strategic locations of WAPs to ensure seamless wireless coverage	Yan Kai
10	Future Expansion Plans	- Discussed the availability of extra switch ports and patch cords for scalability	Chiu Xian

Meeting Minutes #2

Date/Time	30th December 2024 7:00pm	
Location	Online (Google Meet)	
Agenda	1. Finalising network cable types and installation plan	
	2. Reviewing the setup of the hybrid classroom and embedded lab	
	3. Discussion on connectivity in staff office and break room	
	4. Planning for future network expansion and scalability	
Meeting Host	Lau Yan Kai	
Attendance		
Name	**Time**	**Reason For Absence**
Elijah She Yu Sheng	7:00pm	-
Lau Yan Kai	7:00pm	-
Brendan Chia Yan Fei	7:00pm	-
Chew Chiu Xian	7:00pm	-
Minutes		

No.	Item Discussed	Details	Person-In-Charge
1	Cable Types and Installation Plan	- Confirmed use of CAT6 for internal connections and fibre optic for ISP links	Elijah
2	Hybrid Classroom and Embedded Lab Setup	- Reviewed advanced technology and equipment requirements for learning	Brendan
3	Staff Office and Break Room Connectivity	- Ensured reliable network access with appropriate WAP and switch configurations	Yan Kai
4	Future Expansion Plans	- Discussed extra ports and patch cords availability for scalability	Chiu Xian

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